

Divergent Gaze Strategies in Decisions from Experience and Description

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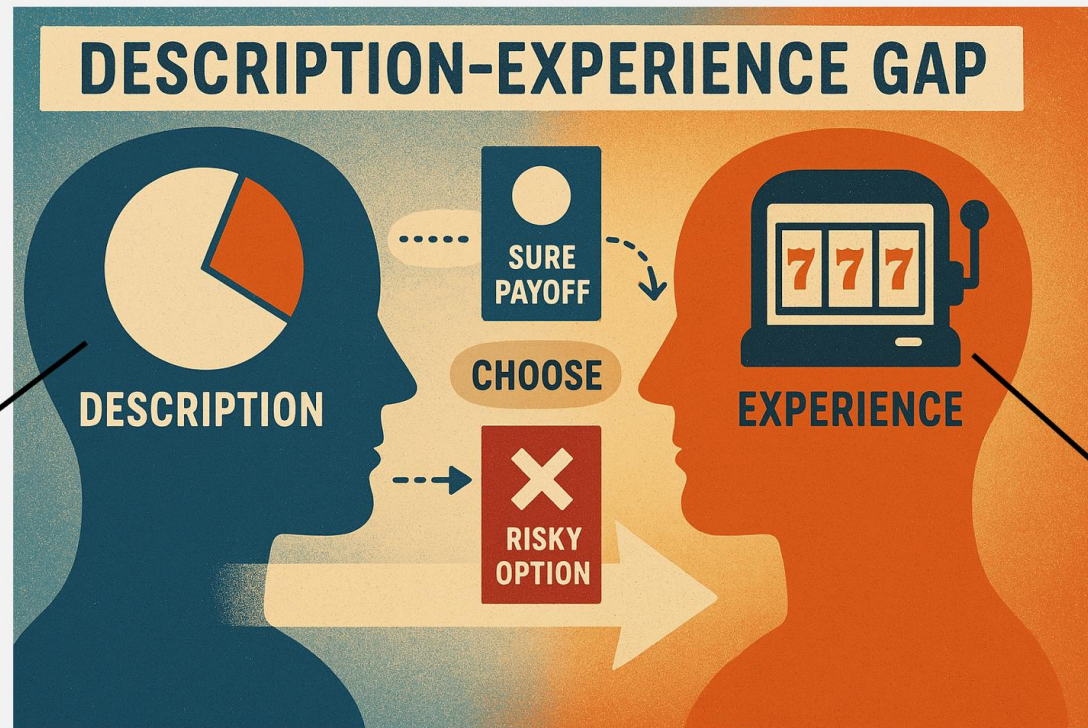


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Background

Description-Experience Gap

- people evaluate risky options differently depending on how the information is obtained, e.g. whether probabilities are explicitly described or learned through experience (Wulff et al., 2018; Barron & Erev, 2003)
- the two formats elicit distinct biases, e.g. core predictions of prospect theory hold under description but break down in experience-based tasks

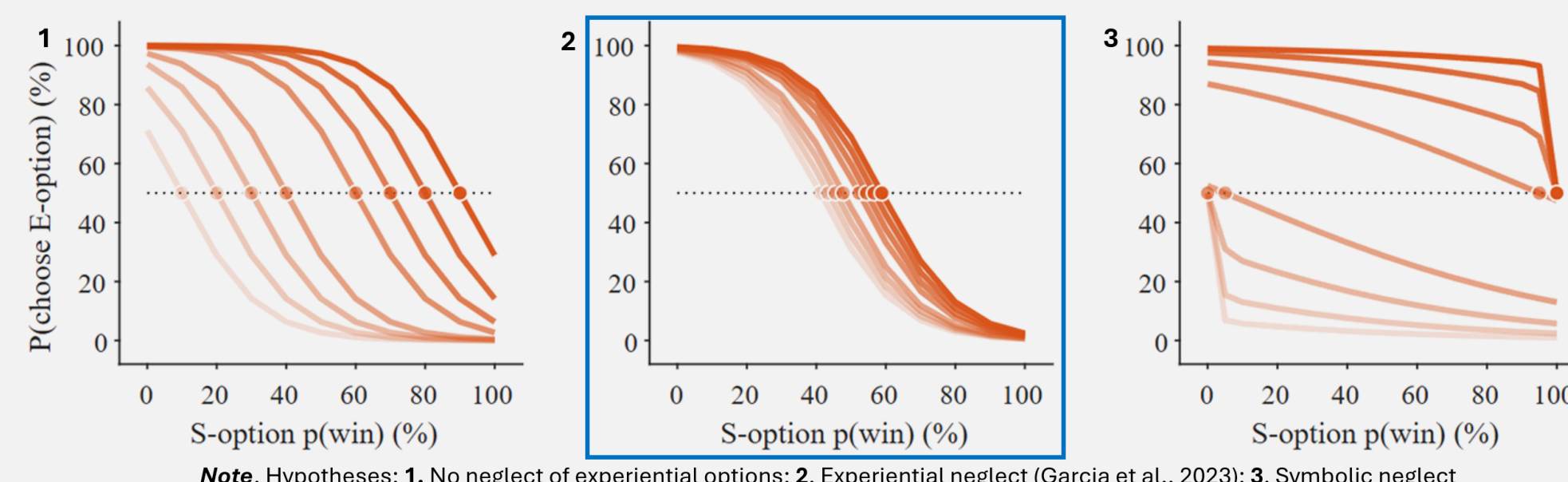


Rare events are typically overweighted when described

Rare events are underweighted when learned from experience (Barron & Erev, 2003; Madan et al., 2019)

The Common Currency Debate

- Diverse types of values are directly comparable in a **unified system for diverse decisions** (Levy & Glimcher, 2012)
- Garcia et al. (2023)** examined hybrid choice tasks where participants chose between experiential and symbolic probabilities
- Participants systematically neglected experiential values**, favouring symbolic information even at an economic cost → modality-specific valuation mechanisms?
- Limitation:** Online study may have limited experimental control



Note. Hypotheses: 1. No neglect of experiential options; 2. Experiential neglect (Garcia et al., 2023); 3. Symbolic neglect

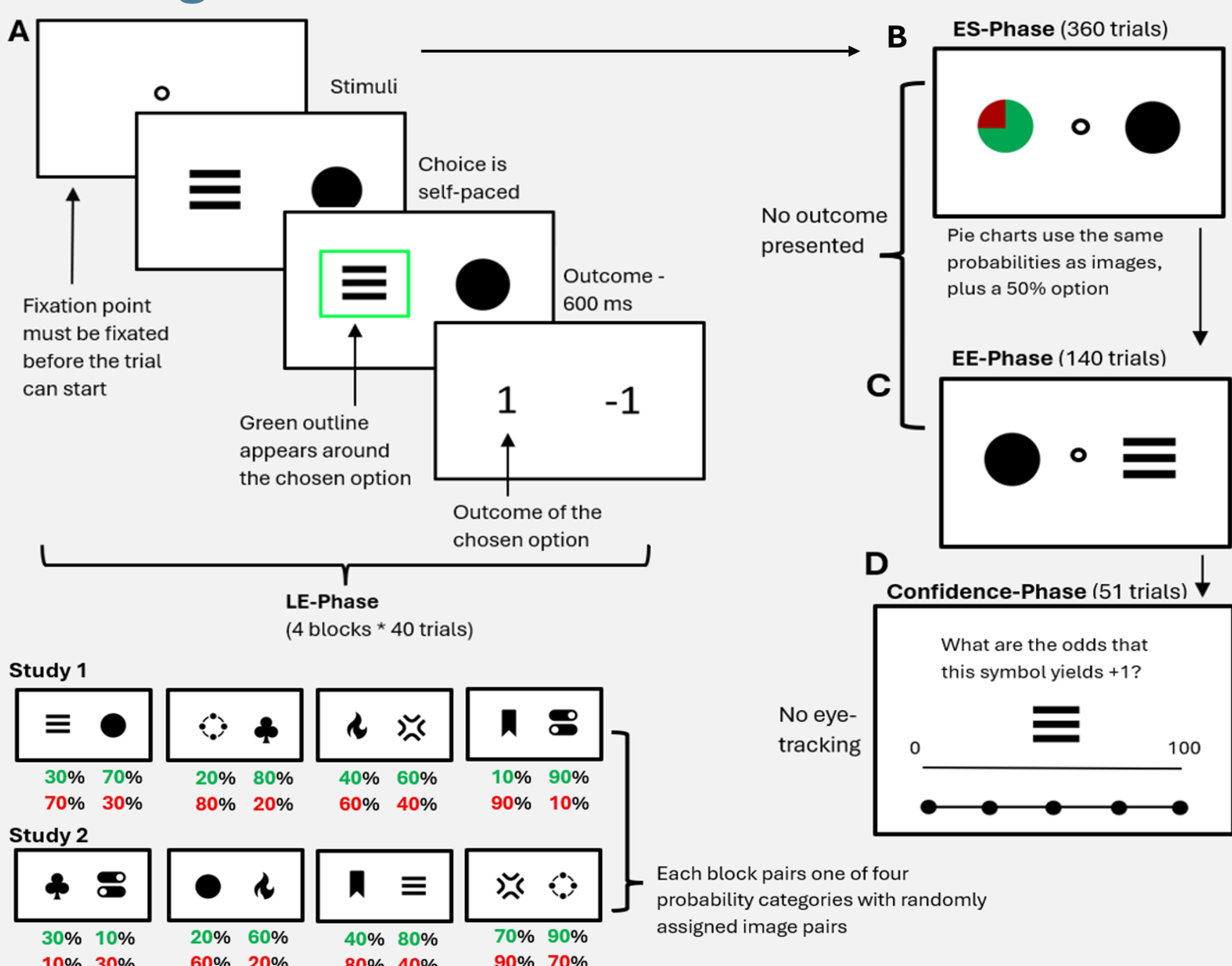
Can we replicate the experiential neglect in a laboratory environment?

- Why does this effect occur, and what are the cognitive mechanisms?
- Can process-tracing techniques help identify the underlying decision dynamics?

We address this combining eye-tracking with the attentional drift diffusion model (aDDM; Krajbich et al., 2010) similarly to **Ting and Gluth (2025)**

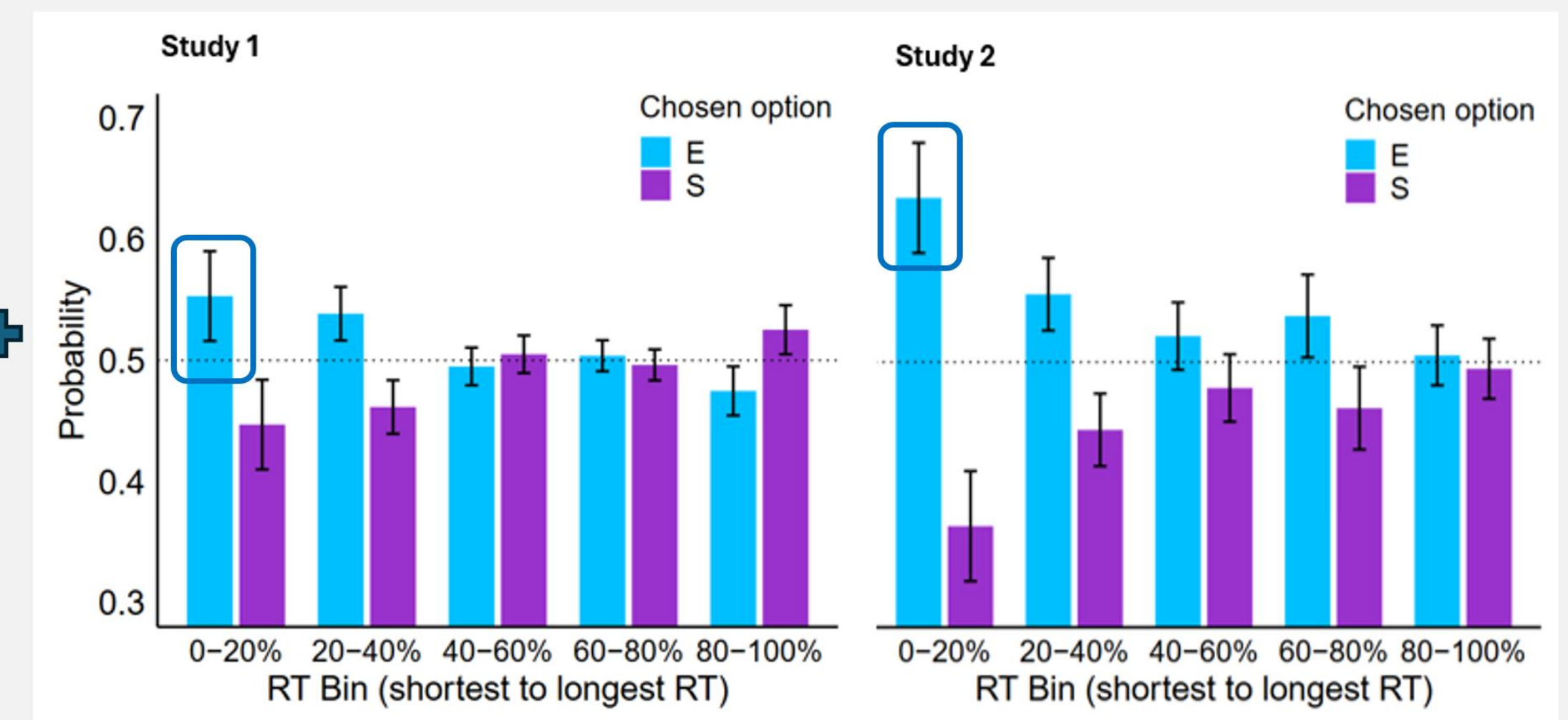
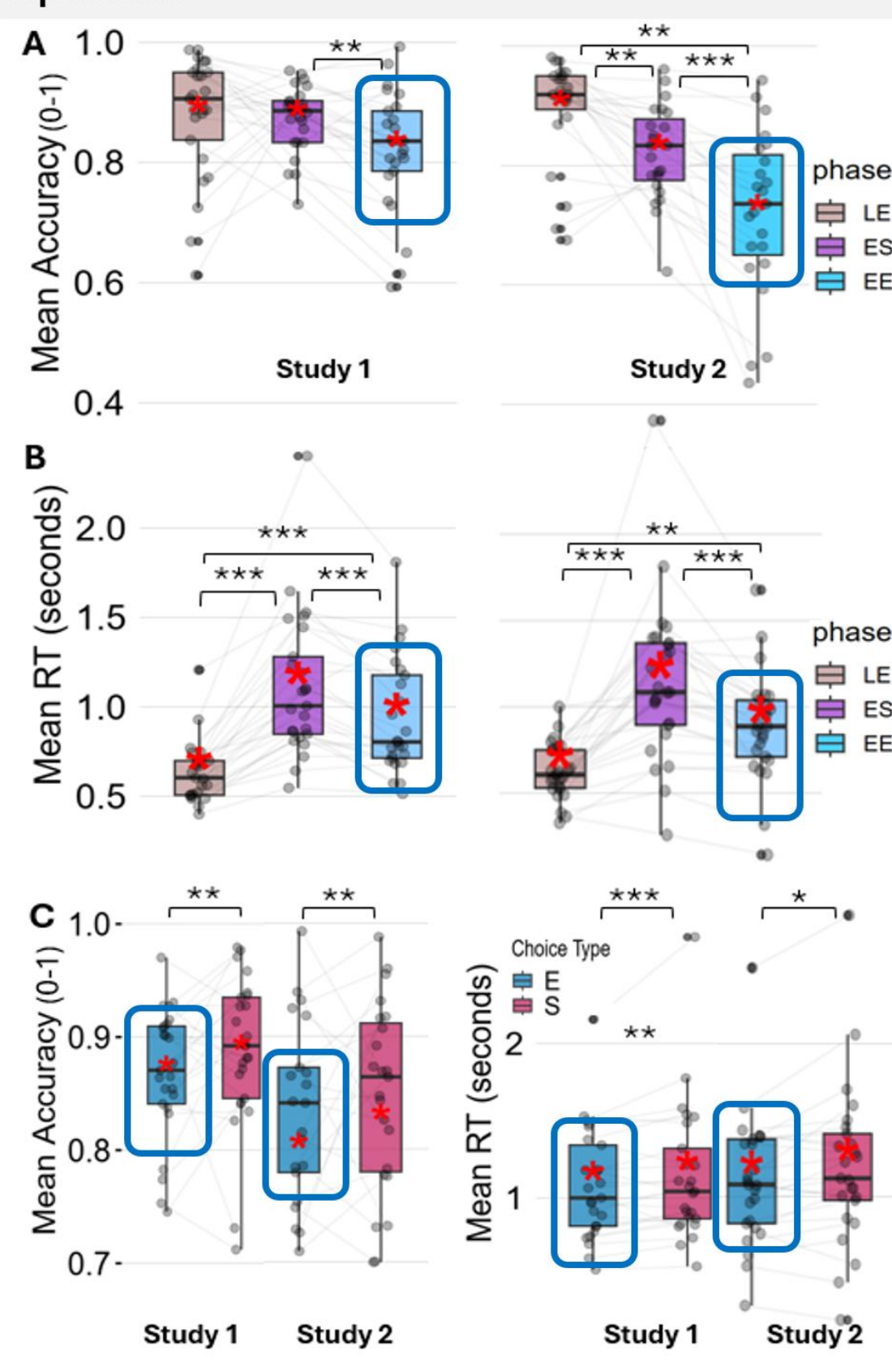
→ We conducted 2 lab studies with 41 participants for eye-tracking (Study 1: $M_{age} = 24$, $SD = 4$, 9 female; Study 2: $M_{age} = 23$, $SD = 3$; 13 female)

Design



Behavioural Results

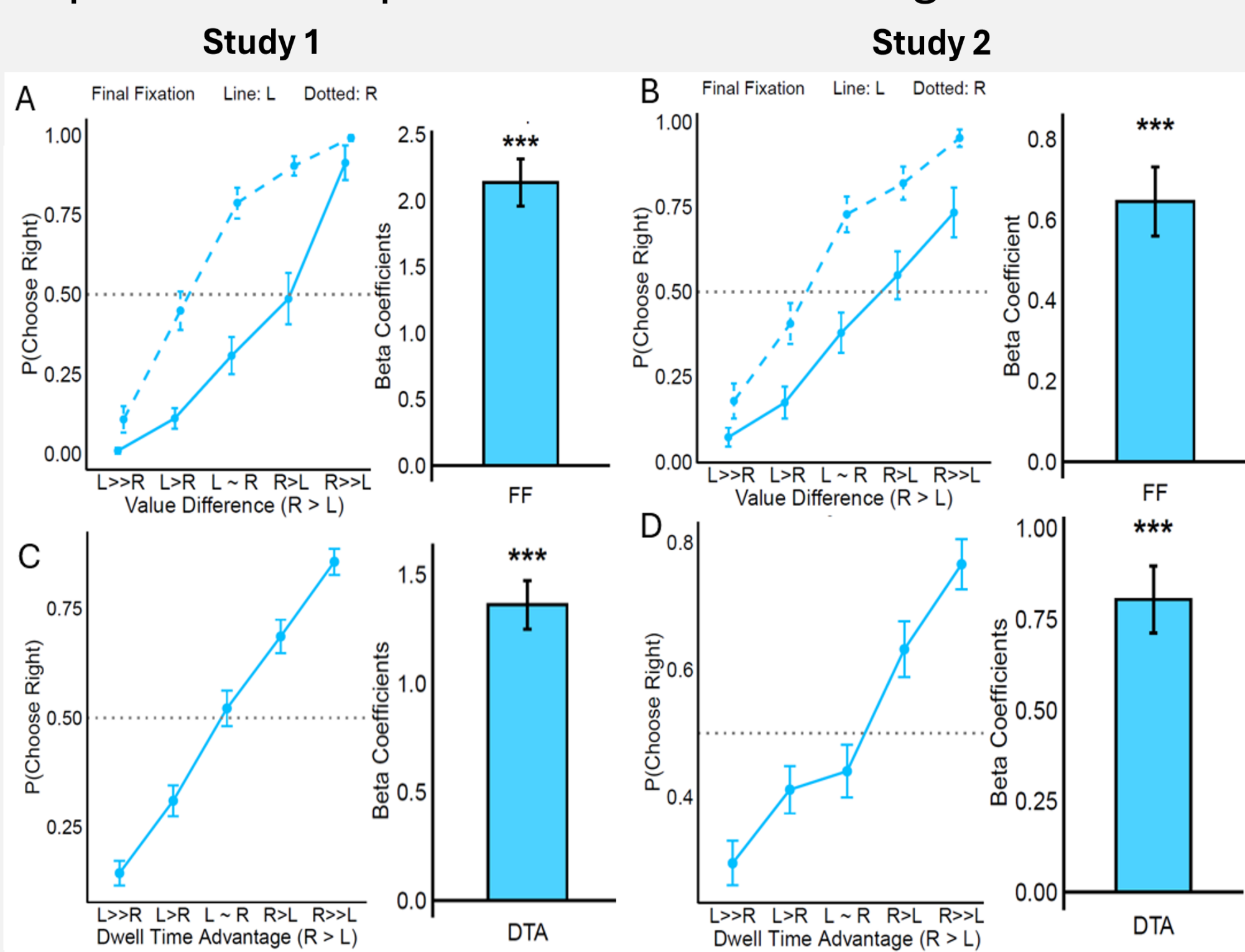
Choices for experiential options are **faster and less accurate**, the fastest choices go towards experiential options



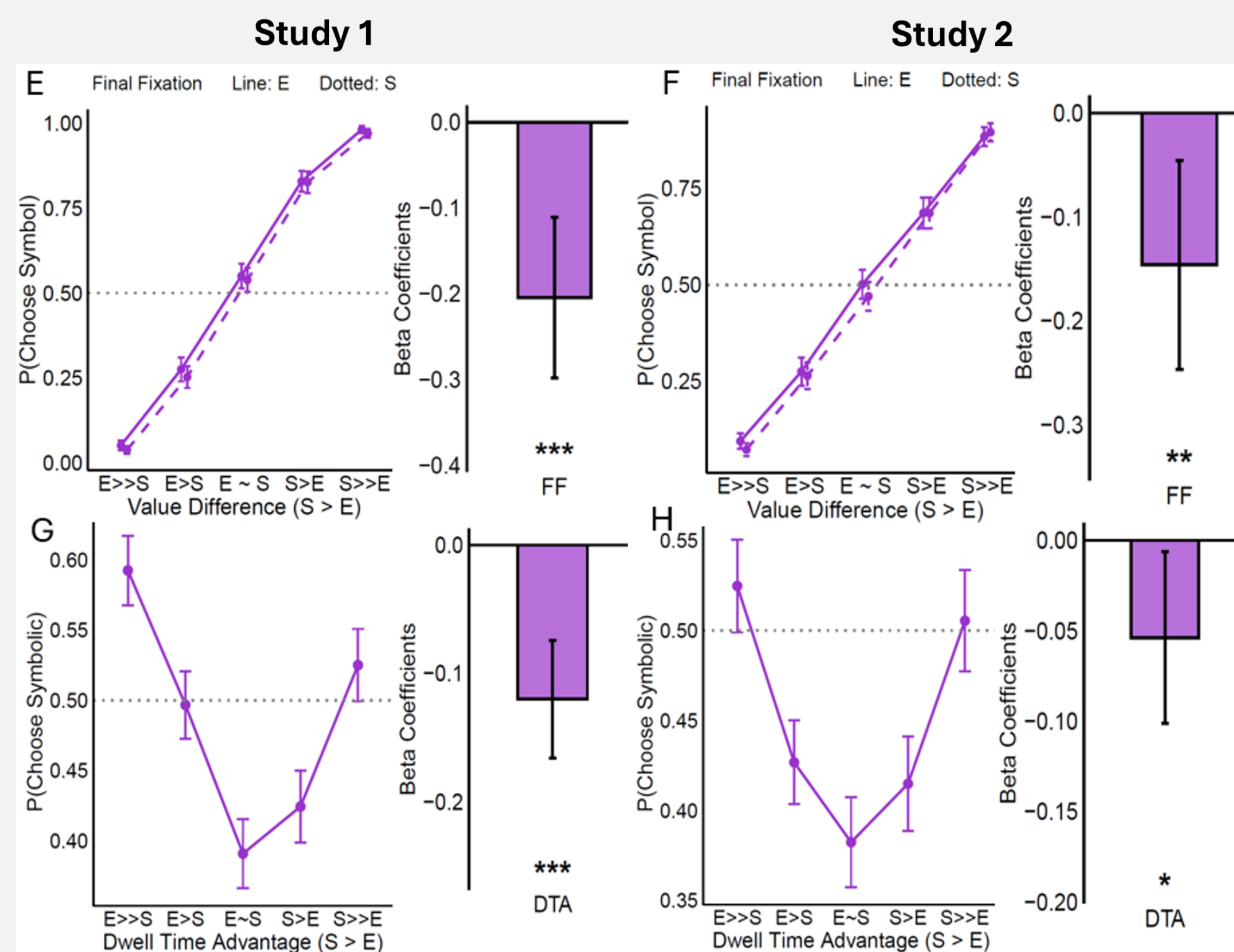
Eye-Tracking Results

Experiential – Experiential Choices → aligned with aDDM predictions

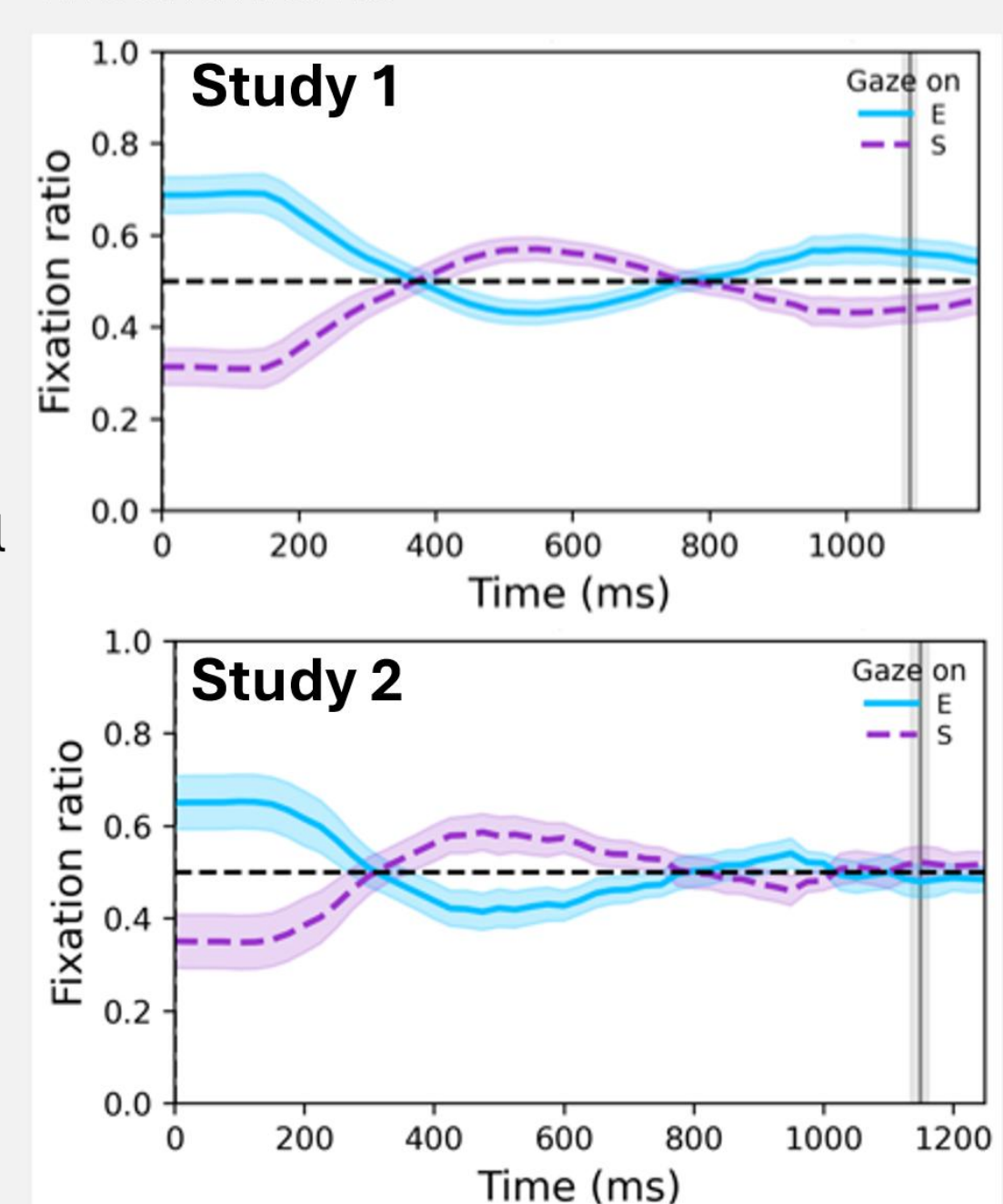
Experiential – Symbolic Choices → surprising U-Shape



- ✓ Final Fixations (FF) predict choice
- ✓ Dwell time on an option predicts choice for that option

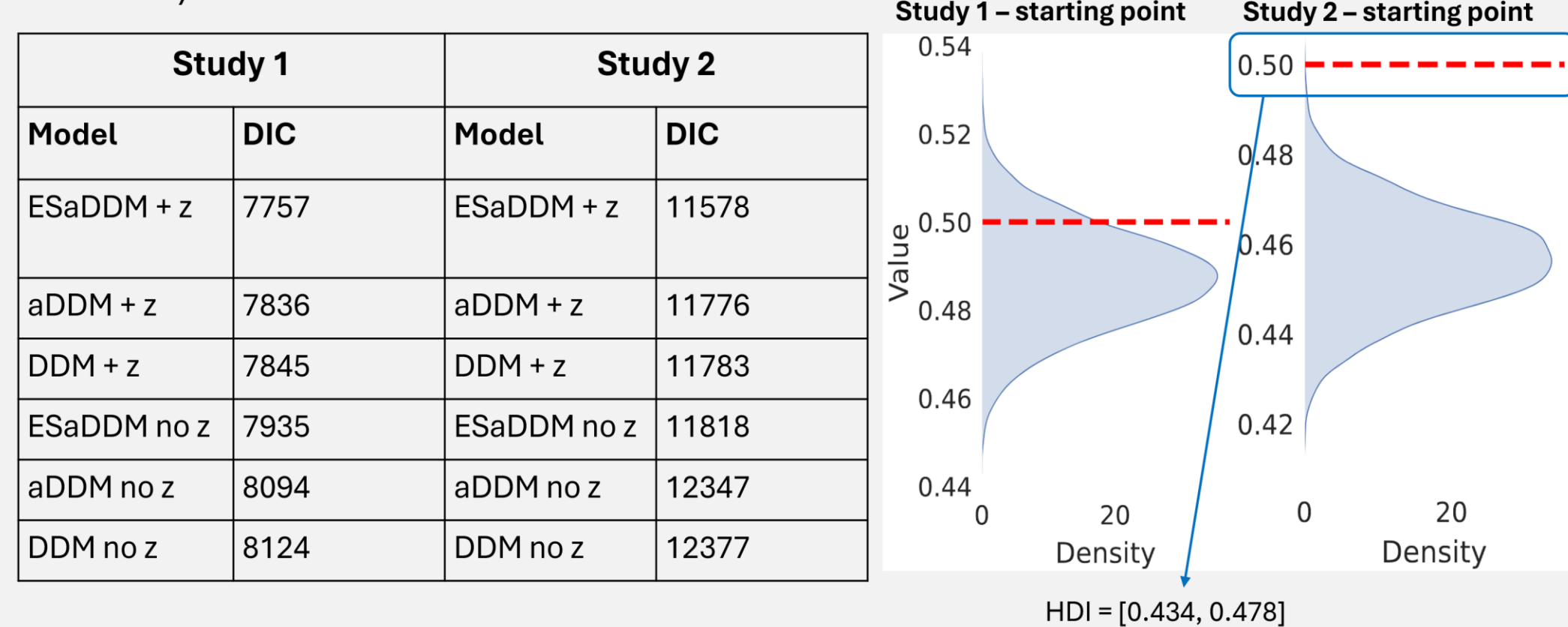


Sliding window analysis shows **systematic & strategic gaze mechanisms**



ESaDDM

- Can our curve be modelled by an aDDM with a starting point bias toward the option from memory?
- Best fitting model: ESaDDM + z (= 2 inattentional weights + starting point bias)



Conclusion

- We fail to replicate the “Experiential Neglect”
- Used an algorithmic-mechanistic approach to understanding this decision trade-off
- Found a novel & systematic gaze-effect that is likely linked to a memory mechanism and confidence
- Gaze-curves are inconsistent with predictions of the aDDM → since our data points to a fixation strategy for optimal choice, should we incorporate this in a model? (using BayesFlow etc. ~ approx. $p(\text{choice}, RT, \text{gaze} | \text{parameters})$)

→ future studies could investigate whether this effect is linked to monitoring and alertness; e.g., P3b ERP related to mental effort (Frömer et al., 2021) and whether it is linked to real-world phenomena such as climate uncertainty

Contact



<https://github.com/VeronikaWendler>

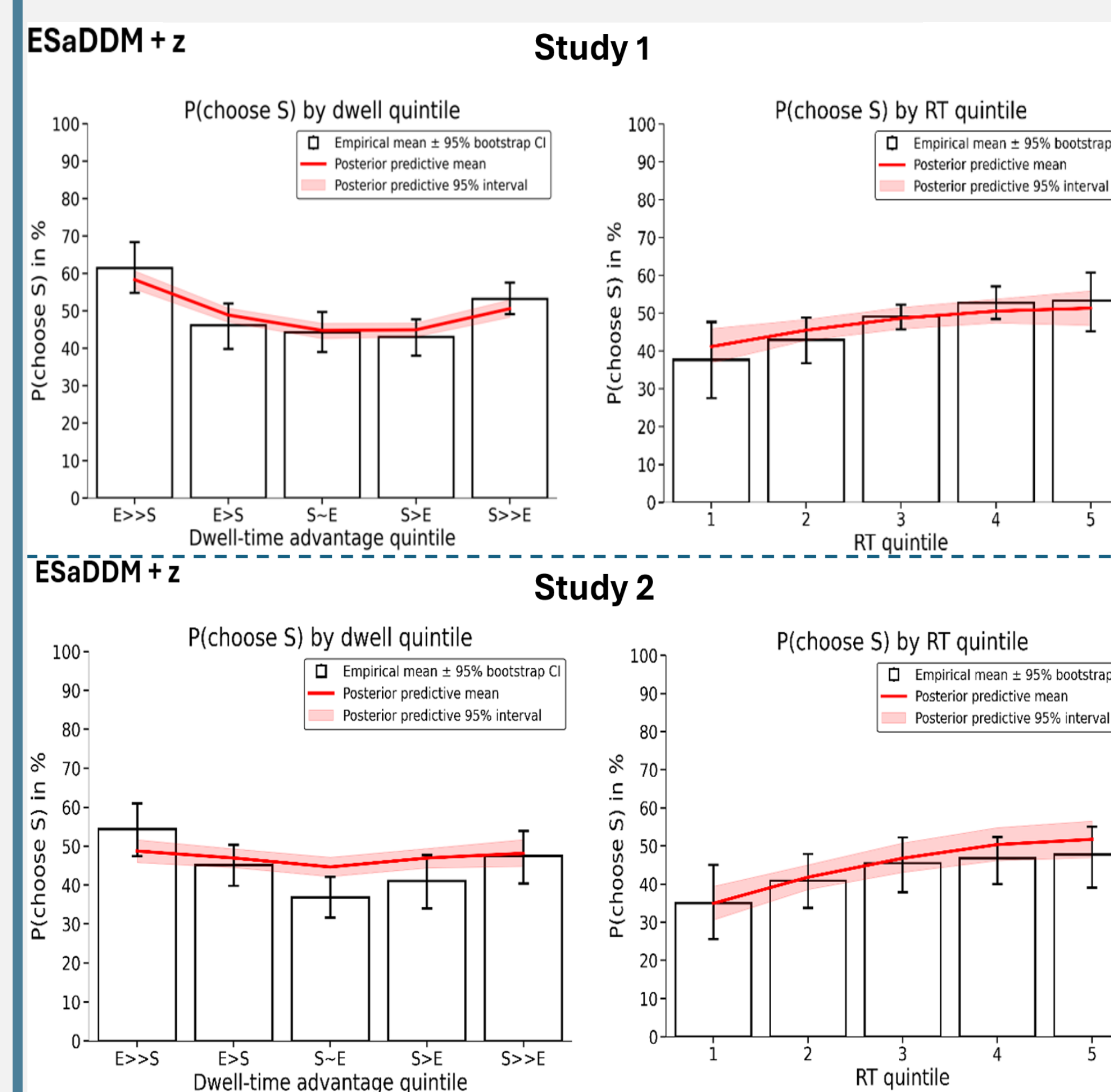


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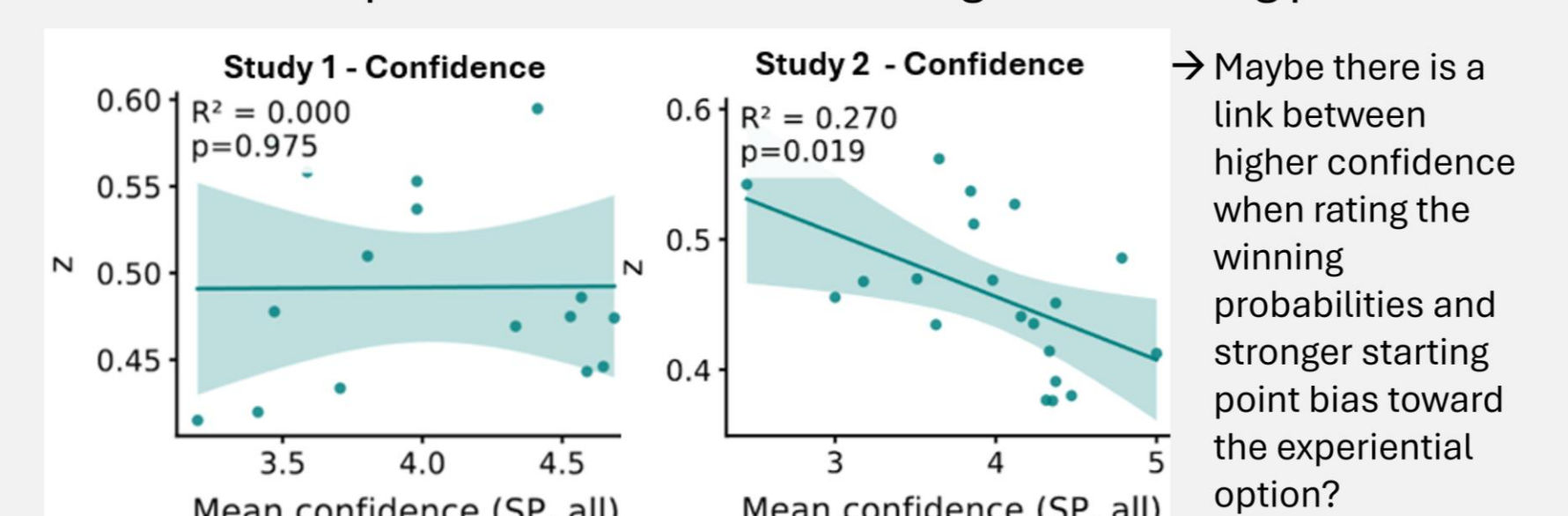
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Posterior Predictive Checks

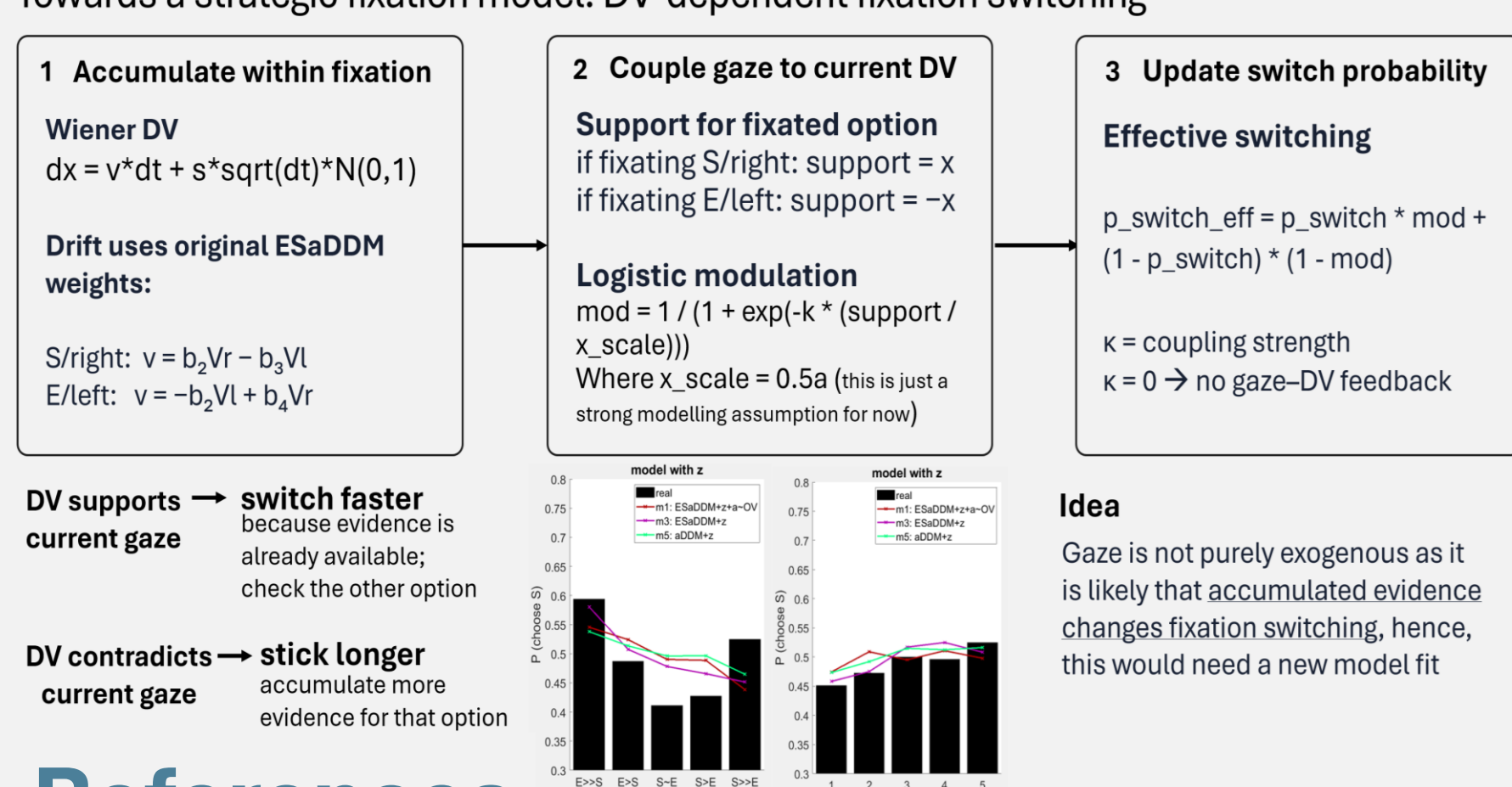


Future Ideas

The relationship between confidence ratings and starting point bias



Towards a strategic fixation model. DV-dependent fixation switching



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